

Think Shallow, Solve Deep: Controlling Recurrent Dynamics for Reliable Test-Time Depth

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Abstract

Recurrent-depth reasoners aim to solve harder problems by iterating their update longer at test time, but additional iterations can improve, preserve, or degrade an answer. We show that a measurable property of the trained operator, its finite-time dynamical regime (estimated as settling, marginal, or drifting), indicates which of these occurs. We give a sufficient condition for depth-safety: once an operator’s per-step displacement is small relative to the decoder margin, the decoded answer cannot change under further iterations. Empirically, on algorithmic tasks trained from 800 unaugmented examples per difficulty tier, settling operators do not degrade with added depth, and on some tasks convert it into higher accuracy on harder unseen instances (Sudoku, 0.19 to 0.34 past the training horizon). A single terminal fixed-point objective moves the regime and the depth behavior together: removing it induces drift and removes the gains, and adding it to a generic recurrence yields depth-safe extrapolation on carry propagation. We give four operational criteria for useful test-time depth, use them to catalogue failure modes, and, as a consistency check, apply the same measurements to Huginn-3.5B, which falls in the non-settling family.

1 Introduction

Recurrent-depth reasoners, networks that apply the same block repeatedly to “think longer” on harder inputs, are an increasingly popular answer to how small models might reason. Yet the Tiny Recursive Model (Jolicoeur-Martineau 2025) solves Sudoku only after ~ 1000 -fold augmentation (without it, the same recurrence scores 0.00 under our protocol); HRM (Wang et al. 2025) depends on augmentation similarly; and at 3.5B parameters, Huginn’s extra recurrence yields only marginal gains that quickly plateau (Geiping et al. 2025; Lu et al. 2025). In all three the architecture has a loop, but training does not make it behave like an algorithm: added iterations do not reliably improve accuracy on harder instances.

A recurrent reasoner is an iterated map $z_{t+1} = f(z_t, x)$, and whether iterating longer helps is a property of that map’s finite-time dynamics: whether the state settles toward an input-dependent fixed point, stays marginal, or diverges. A rich theory of such maps exists (Sec. 2), yet reasoning architectures are designed, trained, and scaled without measuring any of it, compensating with augmentation and parameters. We make that dynamical regime the object of study: we measure it on every trained operator, ask what it predicts about

test-time depth, and intervene on it directly.

Our experiments cover a leave-one-out ablation of a controlled-recurrence recipe (CoRE) and seven re-implemented architectures, all trained under one protocol: fixed, unaugmented data, 800 examples per tier, three seeds (three headline arms re-seeded to $n \approx 20$). The difficulty ladders come in two kinds (Sec. 3.1): ladders that enlarge the input, where success turns on size-invariance rather than dynamics, and fixed-size ladders that only deepen the computation.

On the latter the regime predicts the effect of additional iterations: settling operators convert it (Sudoku 0.19 $\rightarrow$ 0.34) or hold their never-trained plateau (0.92 on reachability) while marginal and drifting ones stay flat, collapse, or decay; which architecture settles is task-dependent (on carry it is a TRM-style attention recurrence, not CoRE); and one terminal objective term switches the regime in both directions (Fig. 2, Fig. 3).

Our contributions:

1. **An operational account of useful test-time depth:** four criteria with measurable failure cases, and a proposition fixing what measured settling guarantees (depth-safety, and a bound on adaptive-stabilization time) and what stays empirical (Sec. 5). A measurement suite logs spectral, dynamical, information-flow, and topological signals for every run.
2. **The dynamical regime predicts the fate of test-time depth:** settling operators are depth-safe and, on some tasks, convert depth, while marginal and drifting operators fail one or both; across our campaign each failure falls into one of four measurable modes, and the spectral and topological signatures agree run-by-run (Secs. 5, 6).
3. **One training term changes both the regime and depth behavior:** the terminal fixed-point objective, removed, turns settling into drift and removes conversion or safety with the architecture unchanged; added to an alternative recurrence, it produces both where added depth had previously degraded the model (Sec. 5.4). We apply the same instruments to Huginn-3.5B (Sec. 7), where they place it in the non-settling family, and a label-free objective derived from the same taxonomy repairs its depth-safety (digit copying 0.00 $\rightarrow$ 1.00) without installing conversion.

2 Related Work

Recurrent-depth reasoners. TRM (Jolicoeur-Martineau 2025) and HRM (Wang et al. 2025) solve hard puzzles with tiny recurrent cores but rely on $\sim 1000\times$ augmentation; Huginn-3.5B (Geiping et al. 2025) scales latent recurrence to 3.5B. Input recall (Bansal et al. 2022), easy-to-hard iteration (Schwarzschild et al. 2021), and constraint-graph message passing (Palm, Paquet, and Winther 2018) supply recipe components (cf. Esteves, Vasco, and Melo 2024; Knutson et al. 2024). Equilibrium Reasoners (Huang, Geng, and Kolter 2026) hypothesize reasoning arises from learned attractors; we measure who learns one and flip it causally. Kohli et al. (2026) see grokking-like transitions behaviorally (cf. Chen 2026; Zhu et al. 2025), and Lu et al. (2025) little latent chain-of-thought in Huginn; we supply the instruments that see both on the inference map. Closest, Labovich (2026) analyzes fixed-point stability of looped transformers theoretically; we measure the trained loop’s dynamics per run, across architectures, and switch them causally.

Dynamics of iterated maps in deep learning. Dynamical isometry (Saxe, McClelland, and Ganguli 2014; Pennington, Schoenholz, and Ganguli 2017), deep equilibrium models (Bai, Kolter, and Koltun 2019), and non-normal transient dynamics in RNNs (Kerg et al. 2019; Orhan and Pitkow 2020; Tarnowski 2020) supply the map-level theory. Anil et al. (2022) showed that path independence correlates with upward generalization in equilibrium models and requires weight tying and input injection, corroborating our recall-ablation collapse; our R4 (Sec. 5.1) is its single-initialization face. Our delta: a failure taxonomy across the full ablation $\times$ architecture matrix, a training-objective switch with two distinguishable consequences (conversion vs. depth-safety), and the spectral-anisotropy separation of settled-into-an-algorithm from merely-settled.

Adaptive halting has its own lineage (Graves 2016; Banino, Balaguer, and Blundell 2021); our R2 differs in demanding that adaptivity convert to correctness (Sec. 5.1). Edge-of-stability (Cohen et al. 2021) concerns the training-loss Hessian, not the inference map. A line of work imposes a dynamical property a priori: Bai, Koltun, and Kolter (2021) stabilize DEQ equilibria by regularizing the update’s Jacobian, and recent reasoners follow, including enforced convergence to stable equilibria (Es’kin and Smorkalov 2026), spectral-radius regularization (Yang et al. 2026), and stability-constrained adaptive depth (Movahedi et al. 2026). We instead measure which properties are learned, show that $\sigma_{\max}$ -control specifically fails (Sec. 6), and explain ρ -control’s success from the non-normal structure of trained operators.

Delayed generalization and progress measures. On small algorithmic datasets, generalization can arrive long after the training loss saturates (Power et al. 2022), the generalizing circuit forming gradually, trackable by hidden progress measures before behavior (Nanda et al. 2023; Barak et al. 2022), in competition with a cheaper memorizing circuit (Varma et al. 2023). Our algorithm-versus-coverage question transfers this competition to the inference map; Sec. 6.2 reports which quantities behave as progress measures for recurrent reasoners.

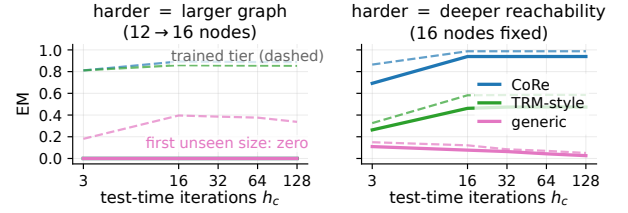


Figure 1: Two notions of “harder,” one task family. *Left*: larger graphs, competent on trained sizes (dashed), transfer nothing. *Right*: same-size, deeper instances, where settling models generalize and plateau.

Finally, persistent homology and topology divergence (Barannikov 1994; Barannikov et al. 2022) ask whether latent trajectories loop, drift, or settle; we test three of its hypotheses at campaign scale (Sec. 6).

3 Setup and Methodology

3.1 Problem statement and the axis we isolate

A weight-tied reasoner is an iterated map $z_{t+1} = f_{\theta}(z_t, x)$ trained for T_{train} iterations on a fixed distribution of easier instances. We ask: when does increasing the test-time iteration budget $h_c > T_{\text{train}}$ (i) preserve acquired competence and (ii) convert into accuracy on harder, out-of-distribution instances?

Two readings of “harder” must be separated; the question above is well-posed under only one. On a fixed-input-size ladder only the number of rounds a reference algorithm needs grows, so extra iterations are exactly the demanded resource; on a size/length ladder success is instead governed by the operator’s invariance to input size, a property orthogonal to its dynamics. Figure 1 makes the distinction concrete on one task family. All extrapolation claims therefore live on fixed-input-size ladders; size/length generalization appears only as scope.

3.2 Tasks, difficulty ladders, fixed data

Three fixed-input-size ladders with deterministic oracle-round counts carry the primary experiments:

- **Sudoku:** 9×9 ; hole fraction $0.2 \rightarrow 0.7$, training tiers $0.2/0.35/0.5$;
- **reachability:** 16 nodes; exact BFS depth $2 \rightarrow 12$, reachable fraction constant by construction;
- **addition:** 2×32 bits, interleaved; longest carry run $2 \rightarrow 28$.

Maze/sorting ladders probe the boundary; size/length ladders support Sec. 3.1; ARC-1/2 (transductive, no augmentation (Chollet 2019)) is supporting only. Training uses *easy tiers only* (rounds well below T_{train}), 800 unique unaugmented examples per tier; evaluation covers the full ladder (256 fixed per tier); splits are materialized once, shared byte-for-byte; three seeds per configuration.

One run = one trained checkpoint per task $\times$ configuration $\times$ seed (evaluations over tiers and depths are repeated observations; parity is excluded as chance-level for every arm). The headline aggregate is

exact-match accuracy (EM; the fraction of complete outputs satisfying the correctness rule below) on the *first unseen difficulty tier*, at the training depth and at the best test-time depth (Table 1); means over all unseen tiers are lower and never mixed with it.

3.3 Metrics: validity oracle and exact match

Sudoku puzzles with many holes admit multiple valid solutions, so exact match against the generator’s reference solution is ill-posed. We score a prediction correct if and only if it is a *valid* completion of the input (validity oracle), alongside reference match em_{ref} . A solution counter quantifies the ill-posedness: the uniquely-solvable fraction per tier is 1.00/0.91/0.50/0.06/0.00; on unseen-hard Sudoku the valid/reference gap is $5.5 \times$ (0.34 vs. 0.062; Sec. 5 confronts the surplus). Aggregates are per-puzzle exact match (never per-cell, which saturates misleadingly) and *solve steps* (the first iteration at which the decoded answer stops changing); on reachability and addition the oracle-round count is deterministic from the input, so adaptivity is checked against required rounds directly, free of Sudoku’s caveat.

3.4 Protocol, test-time depth, architectures

All ladder models train at 16 recurrent steps with deep supervision (4 supervised unrolls), AdamW, gradient clipping 0.5, batch 64, 4–8k steps (task-dependent), $d=256$; at evaluation we sweep test-time depth $h_c \in \{3, 16, 32, 64, 128\}$, the “think longer” axis (per-task settings in the appendix). We compare *mechanisms*, not checkpoints. Seven architectures (TRM-style (Jolicoeur-Martineau 2025), UT/URM (Dehghani et al. 2019), FPRM (Movahedi et al. 2026), EqR, DEQ (Bai, Kolter, and Koltun 2019), Neural GPU (Kaiser and Sutskever 2016), SE-RRM) are re-implemented at matched size and budget and trained on the identical data files, protocol, seeds, and logging as every ablation arm (not authors’ checkpoints). The one pretrained model, Huginn-3.5B, is measured via forward hooks (measure-only; Sec. 7).

3.5 The measurement toolkit

Every run logs, at the trained model on held-out inputs, five families of signal: *spectral* (step-Jacobian σ_{max} , ρ , Henrici non-normality (Henrici 1962), random-probe gain); *dynamical* (top-8 finite-time Lyapunov spectrum via Benettin QR (Benettin et al. 1980), $\sigma_{\text{eff}} = \exp(\lambda_{\text{max}})$, Kaplan–Yorke dimension, fixed-point residual, settle ratio); *information-flow* (logit-lens rank of the correct answer (Belrose et al. 2023)); *topological* (Ripser (Bauer 2021): Betti bars, winding, effective dimension); and *training-time* ranks (full inventory and estimator scope in the appendix). Regime labels: **settle** ($\lambda_{\text{max}} < -0.05$), **marginal** ($|\lambda_{\text{max}}| \leq 0.05$), **drift** ($\lambda_{\text{max}} > 0.05$).

4 The Recipe, Mechanistically

Each recipe component removes one measured way for the loop to avoid becoming an algorithm; we present each through its ablation failure (Sec. 5).

Input recall: keep the problem in the loop. All models condition on the embedded input once per step; the recipe *re-injects* it before each update (Bansal et al. 2022). Removing only this extra injection collapses the trained operator to an *identity* ($\sigma_{\text{max}} = 1.0000$ to four decimals, EM 0.00): the loop discovers an input-insensitive identity solution; the reinforced injection makes the fixed point strongly *input-dependent*, the minimal requirement for the limit to encode an answer.

Structured message passing: the computation graph is the constraint graph. Our step mixes states along rows, columns, and boxes (task-appropriate neighborhoods elsewhere) (Palm, Paquet, and Winther 2018). Removing it keeps the model able to settle (the no-structure arm contracts hardest of all) and it still scores 0.00: settling is necessary, not sufficient. Structure buys what the loop computes; dynamics control buys that it converges.

Fixed-point objective: make the answer a fixed point. We penalize the displacement of the final update at the supervised horizon, $\|z_T - z_{T-1}\| = a_T \|f(z_{T-1}, x) - z_{T-1}\|$, i.e. the fixed-point residual evaluated at the penultimate state, scaled by the learned update gate. This teaches the map to make its answer a fixed point (the gate does not trivially satisfy the penalty by closing; Appendix J). This is the principal objective term evaluated by ablation: removing only this term leaves the architecture, data, and σ_{max} (≈ 5.6) intact but flips the trained regime from settle to drift ($\lambda_{\text{max}} - 0.06 \rightarrow +0.10$, Kaplan–Yorke dimension $0 \rightarrow 8$), and extrapolation dies ($0.34 \rightarrow 0.03$ on unseen-hard Sudoku): the attractor is what makes “think longer” safe: iterations beyond the horizon approach the answer instead of leaving it (Sec. 5).

Remaining stabilizers and their measured scope. The remaining components survive ablation with their means intact and are claimed only for what they measurably buy (Appendix F, Table 3). Orthogonal initialization (Saxe, McClelland, and Ganguli 2014) leaves accuracy unchanged (0.34 vs. 0.34) but selects a different solution family (weight stable rank $\approx 70\text{--}89$ vs. $20\text{--}30$); the adaptive update magnitude and gradient clipping buy seed stability (extrapolation s.d. 0.001 vs. 0.023; ± 0.01 vs. ± 0.13) at task-dependent cost (reachability trainability 0.99 vs. 0.69); deep supervision the same. One popular component fails scrutiny: a lower learning rate is not essential: the higher-LR ablation outperforms the recipe (0.42 vs. 0.34), so we report LR as “matched to scale” and keep the pre-registered headline. The recipe’s effect comes from recall, structure, and the fixed-point objective.

5 Why Some Networks Learn Recursion, and Some Do Not

5.1 What we mean by learned recursion

We use four operational criteria for whether a trained model makes useful and safe use of test-time depth on a task family. The criteria are behavioral and do not by themselves establish that a particular algorithm was learned; a model meets them if all four hold:

- R1** *Extrapolation by iteration.* On instances harder than anything seen in training, accuracy *increases* with test-time iterations beyond the training depth. (Ruled out: distributional coverage, whose depth curve is flat.)
- R2** *Difficulty-adaptive stabilization that converts.* The suffix-stabilization step (the earliest iteration after which the decoded answer never changes through the evaluation horizon) tracks instance difficulty *and* the stabilized answer is correct. (This is a retrospective diagnostic, not an on-line halting rule. Counterfeit: any non-degenerate map’s latents move longer on harder inputs, so adaptivity alone proves nothing.)
- R3** *Incremental computation.* The correct answer forms gradually across iterations, visible as a monotone descent of its logit-lens rank. (Counterfeit: guess-then-polish, with the answer present at step 1.)
- R4** *Convergent answer.* The *decoded answer* converges: further iterations neither change nor degrade it. The *strong form* is a latent fixed point ($\sigma_{\text{eff}} < 1$, residual $\rightarrow 0$); the *weak form* (answer frozen while the latent moves in directions the readout ignores) is realized by carry-transport solutions (Sec. 5.4); cf. path independence (Anil et al. 2022). (Ruled out: depth-fragility and guess-freezing, the latter formalized by Prop. 1(b), both excluded by conjunction with R1–R3.)

The criteria are deliberately behavioral (none mentions a spectrum or an exponent), while the dynamical quantities of Sec. 3.5 are measured independently of them; that the two coincide run-by-run is this paper’s empirical claim. A model with high accuracy that fails R1–R4 uses recurrence without measurable benefit from added depth: the extra iterations do not contribute, and the solution is typically obtained by coverage (cf. Varma et al. 2023).

One relation among the four is not empirical but forced, and it fixes what “measured settling” can and cannot buy.

Proposition 1 (stabilization dichotomy). *Let $\delta_t = \|z_{t+1} - z_t\|$ be the per-step displacement (the quantity the settle ratio measures), let the decision margin $\mu(z_t)$ be the largest r with the decoder g constant on the open ball $B(z_t, r)$, and let $R_t = \sum_{u \geq t} \delta_u$ be the remaining path length. (a) If $R_t < \mu(z_t)$, then $g(z_s) = g(z_t)$ for all $s \geq t$. (b) If $R_t = \infty$, no finite window of answer-constancy certifies the next step; this can occur even when $\delta_u \rightarrow 0$.*

Proof. (a) For any $s \geq t$, the triangle inequality gives

$$\|z_s - z_t\| \leq \sum_{u=t}^{s-1} \delta_u \leq R_t < \mu(z_t), \quad (1)$$

so z_s lies in the margin ball of z_t and $g(z_s) = g(z_t)$, uniformly in s . (b) The map $f(z) = z - (c, 0, \dots)$ with $g = \text{sign}(z^{(1)})$ and generic c , D has $\delta_t = c > 0$, so $R_t = \infty$, yet the answer holds for $\lfloor D/c \rfloor$ steps and then flips, with D arbitrary; a divergent sum with $\delta_u \rightarrow 0$ (e.g. $\delta_u = 1/u$) behaves likewise. $\square$

The premise of (a) is measured, not assumed: R_t is the tail of the logged displacement, and geometric decay gives an explicit freeze time growing with the transient scale and with

$1/\mu$ (Appendix A). Branch (a) makes R4 depth-safety and the adaptive-timing half of R2 consequences of *measured* settling (while decay persists), not of correctness, so settle-but-wrong remains possible, and both R1 and R2’s conversion requirement stay empirical. Branch (b) is the formal definition of guess-freezing, whose premise Huginn meets (Sec. 7). The rest of Secs. 5–6 measures the premises and consequences of Proposition 1 on small controlled models, and Sec. 7 shows branch (b) at scale.

5.2 Test-time depth across ladders and architectures

Across the three primary ladders, three regularities emerge in the depth curves (Table 1, Figure 2; each run colored only by measured regime, boundary dotted, the appendix aggregates by regime).

Learnability is not the bottleneck: most arms fit training ($\text{EM} \geq 0.97$ in 32/51 cells of Table 3); rows below the competence gate ($\text{EM} < 0.3$) are excluded from extrapolation verdicts.

Conversion tracks the regime, not the architecture: on Sudoku the settling recipe family reaches 0.34 ± 0.04 (variants up to 0.42; pre-registered headline kept); on reachability (whose ladder tops out below the horizon, so beyond-horizon conversion is not in play) the settling family generalizes to never-trained depths (0.92 ± 0.10) on a flat plateau; on carry the settling run is TRM-style (0.82 ± 0.16 : its attention step realizes the carry algorithm our local mixing cannot, its halting time nearly matching the oracle’s carry length), while marginal runs rise then collapse (UT/URM $0.95 @ 32 \rightarrow 0.05 @ 128$). TRM also shows the link within its Sudoku seeds: the settling seeds extrapolate while near-marginal ones do not.

Augmentation cuts both ways: applied to our TRM reimplementation (its native data regime, $2 \times$ budget) it lifts Sudoku to 0.16 and recovers a previously unsuccessful seed, an optimization aid (no claims about TRM’s ceiling); applied to the recipe it leaves training tiers and the settling regime intact but collapses extrapolation ($0.34 \rightarrow 0.05$) while flattening the Lyapunov spectrum (Sec. 6.2): coverage neither needs nor builds the anisotropic operator that extrapolation runs on.

Each ablation damages the property it targets: the fixed-point objective is analyzed causally in Sec. 5.4; removing structure or recall removes trainability itself (reachability $0.99 \rightarrow 0.15$). Across the 304 original-ladder runs, *all 19 extrapolating constraint-propagation runs are non-drift* (λ_{max} median -0.047 vs. $+0.009$, $p = 7 \times 10^{-6}$).

5.3 One task, every way to fail it

The patterns of Table 1 become mechanically transparent on Sudoku, where the target algorithm (constraint propagation) is known. The full recipe passes all four criteria: it climbs from 0.00 (shallow) and 0.19 (training depth) to 0.34/0.11 on unseen tiers 3/4 (R1); solve steps grow monotonically ($2.8 \rightarrow 19.6$), commensurate with oracle rounds where propagation terminates and monotone on the 94%-under-determined hard tiers (R2); rank decreases monotonically

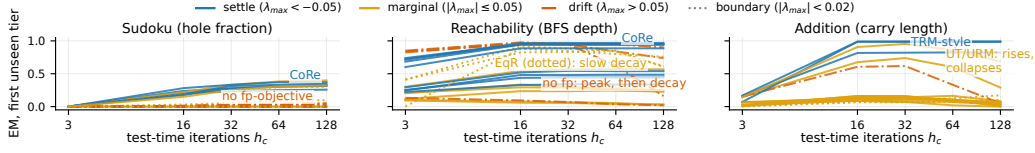


Figure 2: Depth curves behind Table 1: EM on the first unseen tier; one line = one run, colored by *measured* regime (boundary dotted, drift dash-dotted): settling runs rise and plateau, marginal rise and collapse or stay flat, drifting decay here or on deeper tiers (Sec. 5.4); which architecture settles differs per ladder.

	Sudoku (hole fraction)			Reachability (path length)			Addition (carry length)		
	learn $\uparrow$	extrapolate $\uparrow$	overthink $\downarrow$	learn $\uparrow$	extrapolate $\uparrow$	overthink $\downarrow$	learn $\uparrow$	extrapolate $\uparrow$	overthink $\downarrow$
CoRe (full recipe)	1.00	0.19 $\rightarrow$ 0.34 ± 0.04	0	0.98	0.92 $\rightarrow$ 0.92 ± 0.10	0	0.99	0.13 $\rightarrow$ 0.17 ± 0.15	0.05
– structure	0.00	—	n/a	0.15 ± 0.02	—	n/a	= CoRe (no spatial structure on 1-D)		
– extra input injection	0.29 ± 0.01	—	n/a	0.23	—	n/a	0.04 ± 0.02	—	n/a
– fixed-point objective	1.00	0.01 $\rightarrow$ 0.03 ± 0.01	0.04	1.00	0.97 $\rightarrow$ 0.97 ± 0.02	0.37 †	1.00	0.14 $\rightarrow$ 0.15 ± 0.02	0.81 †
generic recurrence	0.00	—	n/a	0.17 ± 0.01	—	n/a	1.00	0.11 $\rightarrow$ 0.12 ± 0.02	0.84 †
TRM-style	0.94 ± 0.18	0.14 $\rightarrow$ 0.21 ± 0.11	0	0.70 ± 0.17	0.55 $\rightarrow$ 0.56 ± 0.22	0.01	1.00	0.80 $\rightarrow$ 0.82 ± 0.15	0.02
UT/URM	0.97 ± 0.06	0.00 $\rightarrow$ 0.00	0.18	0.43 ± 0.09	0.28 $\rightarrow$ 0.28 ± 0.07	0.09	0.99 ± 0.02	0.66 $\rightarrow$ 0.72 ± 0.21	0.59 †
FPRM	0.53 ± 0.40	0.00 $\rightarrow$ 0.01 ± 0.01	0.04	0.31 ± 0.05	0.20 $\rightarrow$ 0.20 ± 0.03	0.11	0.82 ± 0.26	0.27 $\rightarrow$ 0.28 ± 0.25	0.80 †
EqR	1.00	0.01 $\rightarrow$ 0.01 ± 0.01	0.13	0.99 ± 0.01	0.88 $\rightarrow$ 0.89 ± 0.03	0.24 †	1.00	0.12 $\rightarrow$ 0.12 ± 0.03	0.72 †
DEQ	1.00	0.00 $\rightarrow$ 0.00	0	0.26 ± 0.04	—	n/a	0.99 ± 0.01	0.19 $\rightarrow$ 0.47 ± 0.19	0.07
Neural GPU	0.00	—	n/a	0.76 ± 0.08	0.54 $\rightarrow$ 0.54 ± 0.11	0.68 †	—	—	—

Table 1: Three questions, kept separate (mean $\pm$ sd): the primary ablations and all architectures (Appendix F, Table 3 for six further ablations, none switching mode). **learn** $\uparrow$: in-distribution EM at best depth. **extrapolate** $\uparrow$: first-unseen-tier EM at training depth $\rightarrow$ best over $h_c \leq 128$, selected per run on the evaluation set (R1 conversion). **Bold** = best value per column, ties included (max for learn/extrapolate, min for overthink; gated cells excluded). **overthink** $\downarrow$: worst over tiers of best EM – EM@128 († = severe ≥ 0.2 loss, undesirable). EM <0.3 fails the competence gate: extrapolate/overthink are n/a (nothing to convert or lose). Reachability’s rounds stay below the training depth, so its middle column is generalization+safety, not conversion; “—” marks undefined or gated cells. Naming per Sec. 4. CoRe, –fixed-point objective, and TRM-style cells carry $n \approx 20$ seeds, architecture rows $n=10$, the remaining ablation rows $n=3$.

cally toward zero (0.33 $\rightarrow$ 0.05; R3); and all eight estimated leading exponents are negative (strong-form R4).

Is a legal completion of an under-determined board still “harder”? It is partly a different competence (constraint satisfaction, not unique deduction), so we split the credit: reference-EM covers essentially every uniquely determined tier-3 puzzle (0.062 vs. 0.059 uniquely solvable), and the validity surplus is not free (every baseline scores 0.00 under the same oracle), arrives through iteration, and recurs on the ambiguity-free reachability ladder.

Every other outcome in the campaign is a specific, measurable violation: *identity* (no-recall: instant convergence to “do nothing”, $\sigma_{\max}=1.0000$, EM 0.00); *marginal* (generic, UT/URM, FPRM: an exactly normal, norm-preserving wander that fits easy tiers, depth curves flat or declining); *drift* (no fixed-point objective: $\lambda_{\max}=+0.10$, extra steps hurt); *settle-but-wrong* (no-structure; TRM on ARC: dynamics right, fixed point wrong, settling *isotropically*, spread ≤ 0.08 , where true learners are anisotropic, the instrument of Sec. 6.2); full numbers in Appendix F. The remaining ablation arms degrade gradually rather than switching mode, the sense in which recall, structure, and the fixed-point objective are the necessary components.

5.4 Depth-safety and overthinking: R4 in both forms

The clearest intervention is Figure 3: removing the terminal fixed-point objective, which jointly moves the regime and the depth behavior (i.e. whether the premise of Prop. 1(a) is met), destroys a different half of the thesis per ladder. On Sudoku it destroys *conversion* (the drifting ablation never rises, 0.34 $\rightarrow$ 0.03); on reachability it leaves the peak (0.97) but destroys safety (the drifting run loses up to 0.37 EM by $h_c=128$ while the settling recipe holds 0.92 $\rightarrow$ 0.92). The intervention also runs *constructively*: adding only this term to the unchanged generic recurrence turns the architecture carry propagation destroys (overthink 0.91) into a depth-safe extrapolator (best 0.88, 18 seeds, damage 0); UT/URM moves the same way on Sudoku (0 $\rightarrow$ 0.16–0.19) and FPRM on carry, while EqR does not respond (it hard-wires this penalty at a weight whose residual never resolves; Appendix K) and DEQ stops fitting training altogether (10/10 seeds), the boundaries of the intervention.

The payoff requires a *latent* attractor, not a frozen output: a control penalizing only the decoded answer’s change (latent free) yields neither conversion nor safety (Sudoku extrapolates to 0.00, latent never settling; reachability peaks then decays, worst-tier damage up to 1.00; Appendix K), and a weight sweep shows depth-safety turning on precisely at the

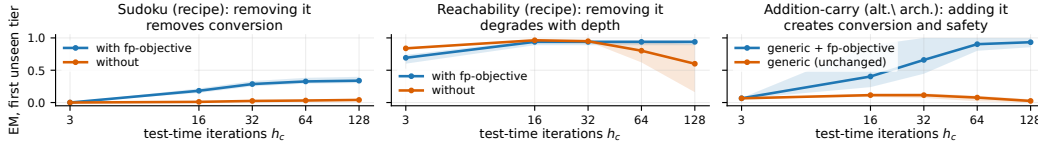


Figure 3: One training term, three consequences (mean and seed range, first unseen tier). Removing the fixed-point objective destroys conversion (Sudoku, left) or safety (reachability, middle); adding only this term to the unchanged generic recurrence creates both on addition-carry (right): $0.12 \rightarrow 0.88$, damage $0.91 \rightarrow 0$.

settling crossing, with conversion growing as the weight rises further; settling is necessary, not sufficient.

Training deeper ($T_{\text{train}}=32$) shows a same-direction gain past its own horizon ($0.43 \pm 0.03 @ 32 \rightarrow 0.48 \pm 0.03 @ 128$; $n=15$ seeds): a longer horizon does not substitute for test-time iteration (pre-registered $T=16$ stays the headline). Without an attractor a model can reach a correct answer transiently but not keep iterating safely: branch (b) of Prop. 1, the operational meaning of overthinking.

R4 is about the answer: a *size-ladder* experiment (off the extrapolation axis, Sec. 3.1) shows why the latent fixed point is R4’s strong, not only, form: on addition-by-length the generic recurrence freezes its answer within ~ 5 steps *while its latent never settles* ($\sigma_{\text{eff}}=1.02$, weak-form R4), yet on prefix sums the same statistics give depth-fragility ($1.00 \rightarrow 0.65$; Appendix I). So latent spectra alone cannot separate weak-form R4 from depth-fragility; the difference lives in answer space, where R4 is stated.

The strong form makes “think longer” a safe default: among all 164 competent runs, not one of the 17 settling runs loses more than 0.004 EM to depth, while 30% marginal and 22% drifting lose more than 0.1, up to 0.98 (Neural GPU on mazes, $p=10^{-3}$). Two notes: under a task $\times$ configuration cluster bootstrap the safety separation survives (excess unsafe 0.14, CI [0.06, 0.24]) while pooled conversion is not cluster-significant (conversion rests on the depth curves and the causal switch), and the ± 0.05 threshold is not load-bearing (Appendix E). On tasks where the recipe fails outright (addition, sort), its own rank curves rise with iterations ($0.74 \rightarrow 0.94$), the information-level image of drift; the same fates reappear behaviorally at 7B (Sec. 7.2).

In sum: learning recursion is passing R1–R4, and across the corpus the runs that do so are consistently settling-but-barely and strongly non-normal, with the settling architecture differing by task.

6 The Shape of Learned Recursion

6.1 Hypotheses and the non-normality account

A topology-first program (Barannikov et al. 2022) predicts settle/loop/drift classes (H1), winding tracking depth (H2), and that a purely contracting map cannot count (H3). Verdicts (Appendix D): **H1** settle/drift ubiquitous and causally separable, loops nowhere; **H2** unsupported (solvers move radially, 99th-pct |winding| 0.71 turn); **H3** supported, with a sharpened mechanism.

Why is σ_{max} a correlate but not a lever (Prop. 2)? Successful operators are strongly non-normal ($\sigma_{\text{max}}/\rho \approx 4\text{--}5.4$): the

top singular value is a *transient* amplification, not an asymptotic rate ($\rho < 1$); random-probe gain stays ≤ 1 , so computation lives in a *thin* amplifying cone inside a globally contracting map. A typical-gain sweep (Bai, Koltun, and Kolter (2021)’s regularizer as a probe) confirms it: pinning the typical gain at its decaying level is harmless; forcing it to expand ($\tau \geq 2$) blows σ_{max} to 14–33 and destroys training (Appendix D). Regularizing ρ (asymptotics only) is the control that helps at scale (Yang et al. 2026); non-normality alone does not (Neural GPU drifts to zero at $\sigma_{\text{max}}/\rho=17.7$).

6.2 What the full corpus adds

Spectral anisotropy: the signature that separates algorithms from mere convergence. The shape of the Lyapunov spectrum is the strongest single predictor we measured: algorithm-learners develop a structured spectrum, a hierarchy of contraction rates (CoRE on Sudoku: spread $\lambda_1 - \lambda_8 = 0.41$), while every failure mode is spectrally degenerate: eight nearly identical exponents, whether pinned at zero (identity), at +0.02 (marginal), or at -0.17 (no-structure: settled hard, isotropically); Appendix C overlays the full landscape.

The spread alone predicts extrapolation with AUC 0.846 across 304 runs, the best single dynamical quantity we measured though on par with training accuracy itself (0.851): a corpus-level correlate whose value is the within-settle separation, not a universal discriminator (reachability’s extrapolators settle at spreads 0.07–0.14).

The Kaplan–Yorke dimension is bimodal (47% exactly 0, 48% saturated), giving R4’s strong form its practical reading: a learner of recursion has all eight estimated leading exponents negative. Within the settling regime the spread separates runs that settled into an algorithm from runs that merely settled (AUC 0.872, $p=10^{-3}$), a distinction no other instrument could make, including on TRM-trained ARC models (settled, flat, wrong); the augmentation control of Sec. 5.2 is the same flattening shown causally. We read this as functional differentiation: an iterated algorithm needs directions with different jobs (one instrument-level exception in Appendix N).

Three further regularities (Appendix N). Live spectra and post-hoc geometry agree run-by-run (σ_{eff} vs. settling $r=0.75$, $n=325$; replicated $\rho=0.69$, $n=1,206$); topology sees compactness, not loops (92/24/2% across regimes; per-instance AUC 0.80–0.98, Remark 1); and the algorithm forms late (rank 33 $\rightarrow$ 53 while siblings stay flat), the inference-map face of delayed generalization (Sec. 2); dash-

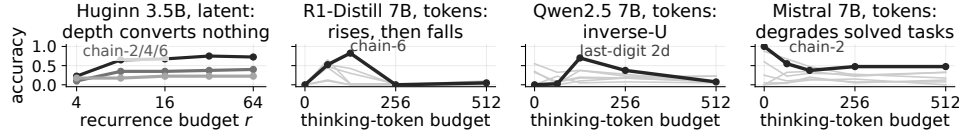


Figure 4: Recurrence at scale, one prompt bank. *Left*: a latent depth knob (Huginn-3.5B) saturates without converting depth. *Right three*: token-budget knobs on open 7B reasoners (bold = illustrative family, $n=40$ /point); none shows the settling fate.

board in Appendix M.

7 Recurrence at Scale

This section tests the taxonomy at scale, then repairs a depth pathology of a pretrained recurrent LLM with a label-free objective. Huginn-3.5B (Geiping et al. 2025) iterates the same block on its latent state; a chain-of-thought reasoner iterates in token space, where only behavior compares; neither is a performance baseline.

7.1 Latent recurrence: Huginn-3.5B

R1/R3 are answered in prior work: recurrence $4 \rightarrow 32$ yields marginal gains far below chain-of-thought (Figure 4, left), no structured latent chain found (Lu et al. 2025).

R4: across per-token latent trajectories on 7 task categories (2,585 tokens), not one settles ($\sigma_{\text{eff}}=1.000 \pm 0.003$; layer normalization pins only the radial direction, not the read displacement): *normalized-marginal*. **R2**: the answer freezes, but freezing does not track difficulty: guess-freezing (Prop. 1b), not a computation completing (Appendix G; Lu et al. 2025); by R1–R4 the loop gains nothing from added depth.

The base’s depth is unsafe, not just inert: on procedural single-token probes accuracy peaks shallow and collapses (carry: 0.69 at $r=8$, 0.00 for $r \geq 32$), every input freezing onto the same wrong token. A settling event itself, guess-freezing evades answer-stability halting; a 30-example depth calibration recovers the peak at the price of labels and a per-task depth the repair below does not need (Appendix H).

Repairing depth-safety without labels. Guided by the taxonomy, a rank-16 LoRA adapter on the recurrent core is trained with one label-free latent objective: pull deep states, $r \sim U[16, 128]$, toward the frozen base’s own shallow state at $r=8$ (two synthetic anchor domains, 2,000 steps); the fixed-point term itself does not transfer to this scale, only its latent-anchoring form (Appendix H). The patch restores depth-safety, not new capability, on tasks never seen in training: digit copying, lost by the base at $r \geq 4$, reaches 1.00 ± 0.00 at $r=128$ with input-dependent answers (10/10 seeds); letter repetition recovers $0.00 \rightarrow 0.51 \pm 0.13$ (a carry plateau: stabilization only; Appendix H). Conversion never appears, and the depth-hard pointer-chase task stays unlearnable under every intervention tried, making the no-benefit-from-depth verdict causal; transfer follows the anchors’ operation family across alphabets, with two failure laws measurable in advance (Appendix H).

7.2 Token recurrence: the same fates

Chain-of-thought gives a reasoner its own depth knob; on one prompt bank three open models reproduce the non-settling fates (Figure 4): the distilled reasoner is destroyed by more of the budget it needs ($0.82 \rightarrow 0.00$) and the non-reasoning model degraded by thinking it does not need ($1.0 \rightarrow 0.47$; a shared failure shape, not mechanism identity; Chen et al. 2024; Su et al. 2025; Sui et al. 2025). Both substrates over-think; the case for latent depth is the handle: only the latent loop exposes a state one can measure and repair (Sec. 7.1).

8 Limitations

Synthetic generators. Sudoku boards come from one construction’s symmetry orbit, so the headline is extrapolation on that generator; the matching reachability plateau shows the claim does not rest on Sudoku’s scoring. **Controlled protocol.** Budget-matched re-implementations estimate no ceiling, which isolates the regime from tuning confounds; producing a settling operator is not guaranteed (on mazes, a seed lottery); where obtained, the outcome follows the regime. **Small scale.** Models are tiny and tasks synthetic, yet the same instruments place Huginn-3.5B in the marginal family; the scale repair (Sec. 7.1) is operation-family-scoped, one checkpoint. **Mediation.** We identify the *objective* as the causal lever and the *latent* attractor as its needed target (answer-only control, Sec. 5.4), not the regime as sole mediator; finite-time estimates bound adjectives, not regime separations (Appendix E).

9 Conclusion

Learned recursion is a dynamical property: a non-normal operator settling in an attractor the objective creates and structure aligns with the task; failures fall into four measurable modes. Depth-safety is provable at small displacement (Prop. 1); conversion stays empirical and task-dependent. The measurements, and the repair they prescribe, carry to 3.5B: whether added depth helps is a measurable property of the dynamics.

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Supplementary Material

A Propositions and proofs

This section gives the geometric-decay corollary of Proposition 1, the trajectory-informativeness remark, and Proposition 2 (all referenced from the main text). Notation as in the main text: $\delta_t = \|z_{t+1} - z_t\|$, decision margin $\mu(z)$, remaining path $R_t = \sum_{u \geq t} \delta_u$; g is the (argmax) decoder.

Corollary 1 (explicit freeze time under geometric decay). If $\delta_u \leq C\gamma^u$ with $\gamma < 1$, then $R_t \leq C\gamma^t/(1-\gamma)$, and by Proposition 1(a) the decoded answer is frozen for all $t \geq t^*$, where

$$t^* = \max\left(0, \left\lceil \log \frac{C}{(1-\gamma)\mu} \bigg/ \log \frac{1}{\gamma} \right\rceil\right),$$

obtained by solving $C\gamma^t/(1-\gamma) < \mu$ (open ball, strict). The freeze time grows with the transient scale C (longer propagation) and with $1/\mu$ (smaller margin, harder instance); with $\gamma = e^{-|\lambda|}$ the leading order is $t^* \sim |\lambda|^{-1} \log(1/(|\lambda|\mu))$, i.e. decreasing in the contraction rate, so this is a monotone heuristic, not an equality. A cross-check on the fixed-size corpus: among settling competent runs, solve steps correlate with $1/|\lambda_{\max}|$ on reachability (Spearman 0.56, $p=10^{-3}$) and weakly, non-significantly on Sudoku/carry, as expected since C and μ also vary per instance and task.

Remark 1 (trajectory informativeness is regime-conditional). In branch (a) the certified freeze time is bounded by a decreasing function of the margin, so per-trajectory statistics encode instance geometry; in branch (b) the freeze step is an entry time into a level set of g and carries no such information. This is a statement about *information*, not *success* (a large margin can surround a confidently wrong state). Empirically, on the settling subset of the fixed-size corpus ($n=132$ trajectory files with both correct and incorrect examples), the correct-minus-incorrect gap in settle ratio, path length, and effective dimension is directional and significant (Wilcoxon $p=3.4 \times 10^{-9}$, 4.5×10^{-7} , 3.9×10^{-11}); on the marginal subset the gaps vanish (medians ≈ 0); and on Huginn (marginal) the freeze step does not separate correct from incorrect within a difficulty rung ($p \approx 0.9$). This is the mechanism behind the per-instance confidence read-out of the main text (Sec. P); the two use different corpora and tests, so their p -values are not comparable.

Proposition 2 (transient/asymptotic decoupling, single linearization). For a fixed step Jacobian J with top right singular vector v ,

$$\|Jv\| = \sigma_{\max}(J), \quad \rho(J) < 1 \Rightarrow \|J^t\| \rightarrow 0 \quad (2)$$

give the one-step amplification and, by Gelfand’s formula, the asymptotic decay. Thus for a non-normal J a large transient gain $\sigma_{\max}$ coexists with asymptotic contraction; clamping $\sigma_{\max}$ removes the transient that carries the computation, whereas constraining ρ leaves it intact. This is elementary and concerns one linearization step; the thin-cone geometry across the trajectory (main text Sec. 6) is measured, not implied by it.

B Analysis populations (run-flow)

Different analyses use different corpora; Table 2 defines each population and the unit of analysis, so counts that differ across the paper (e.g. 17 settling runs in the depth-safety analysis vs. larger settling counts elsewhere) are traceable to distinct campaigns rather than inconsistency. One run = one trained checkpoint (task $\times$ configuration $\times$ seed); tiers, depths, and stored trajectory files are repeated observations within a run, not independent replications.

Corpus	Runs	Used for
Original ladders (LOO + architectures)	304 total; 164 competent (17 settle / 129 marginal / 18 drift)	Depth-safety separation (main Sec. 5) and Fig. 5; anisotropy AUC; non-drift extrapolator count
Fixed-size ladders (primary)	246	Table 1, depth curves (main Fig. 2), headline conversion numbers
Trajectory topology (original)	325 files	$\sigma_{\text{eff}} \leftrightarrow \text{settling}$ ($r=0.75$); per-instance confidence
Trajectory topology (fixed-size)	1,206 files	Rank replication (Spearman $\rho=0.69$); Betti-1 shares

Table 2: Population and unit of analysis for each corpus. Counts differ by design; the depth-safety statistic and the regime figure below both use the original-ladder competent corpus (164 runs).

B.1 Regime-to-outcome aggregate view

Figure 5 aggregates the 164 competent original-ladder runs (the same population as the main-text depth-safety statistic) by measured regime. Settling runs carry the higher median depth conversion, and overthinking damage concentrates in the non-settling regimes; conversion is task-dependent and, pooled, not cluster-significant (main text), so the figure is descriptive of this corpus, not a pooled significance claim.

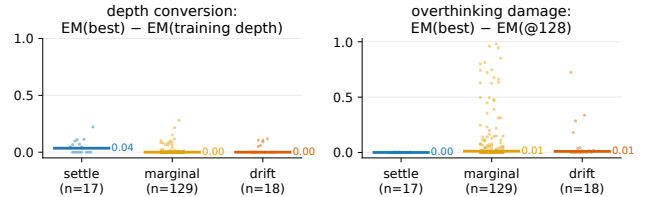


Figure 5: Regime $\rightarrow$ outcome on the 164 competent original-ladder runs (dot = run, bar = median; n per regime in axis labels). Left: depth conversion, $\text{EM}(\text{best}) - \text{EM}$ at training depth on the first unseen tier. Right: overthinking damage, worst over tiers of $\text{EM}(\text{best}) - \text{EM}@128$. Settling runs show no overthinking tail; marginal and drifting runs do.

C The Sudoku ablation landscape in two instruments

Figure 6 overlays every Sudoku arm in the two instruments behind the anisotropy and crystallization claims of the main text (Secs. 5–6).

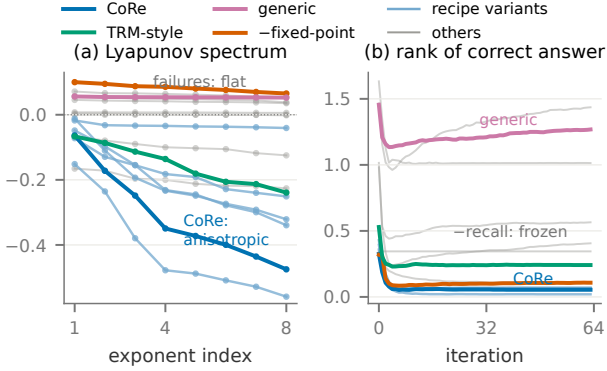


Figure 6: The Sudoku ablation landscape (all arms overlaid; colored = narrative-central operators, light blue = recipe variants, gray = others). (a) Lyapunov spectra: learners are *anisotropic*; every failure family is flat. (b) Logit-lens rank of the correct answer (hardest training tier): CoRe crystallizes to ≈ 0 , generic bounces, the identity ablation is frozen. Depth curves for these arms are in the main text; the training-time formation view is in Sec. O.

D Trajectory-topology hypotheses: full statements

A topology-first research program (Barannikov et al. 2022) predicts that reasoning trajectories fall into settle / loop / drift classes (H1), that winding of the trajectory tracks reasoning depth (H2), and that a purely contracting map cannot implement counting-like computation (H3). Our corpus gives verdicts:

H1: two of three regimes observed; the third is absent.

Settle and drift are ubiquitous and sharply separable (main text Sec. 5), and the regime axis is causal via the fixed-point switch; loops are not observed anywhere, with median total H1 persistence 0.000 in every regime and Koopman/DMD dominant frequencies are ≈ 0 across the corpus.

H2: not supported in this task class. The 99th percentile of $|\text{winding}|$ is 0.71 of a turn, extrapolators included: propagation solvers move radially toward their fixed point, not rotationally (task classes with cyclic state may differ). The same holds at scale: on Huggin-3.5B, $|\text{winding}| > 1$ occurs for only 0.08% of the 2,585 measured token trajectories.

H3: supported, with a sharpened mechanism. The successful operator is *not* globally contracting: it contracts on average ($\sigma_{\text{eff}} < 1$) while strongly expanding along a thin subspace ($\sigma_{\text{max}} \approx 4$ –5), which is where the computation lives (main text Sec. 6). A regularizer sweep that drives the *typical* (random-direction) Jacobian gain toward a target τ (the

Jacobian regularizer of Bai, Koltun, and Kolter (2021), repurposed here as a probe of the learned geometry rather than a stabilizer) confirms this geometry from both sides. Pinning the typical gain at its natural, decaying level ($\tau=0.5$) is harmless: training and extrapolation intact (0.26–0.36), the thin amplifying cone untouched (σ_{max} still ≈ 4), while forcing typical directions to *expand* ($\tau \geq 2$) destabilizes the operator (σ_{max} blows up to 14–33) and destroys training itself. Intermediate targets already degrade seed-wise: at $\tau=1$ one seed in three collapses to 0.004 while the others extrapolate normally (0.37). Random directions must decay; only the thin cone may amplify. Regularizing the spectral *radius* ρ (constraining asymptotics while leaving the transient intact) is likewise the control that helps at scale (Yang et al. 2026).

E Non-normality account and estimator scope: full statements

Why is σ_{max} a correlate but not a lever? Successful operators have $\sigma_{\text{max}}/\rho \approx 4$ –5.4 (Henrici index 0.08–0.10 on Sudoku): they are strongly non-normal, so their top singular value measures a *transient, feedforward* amplification each step applies to fresh input, not an asymptotic growth rate (which is $\rho < 1$). Clamping σ_{max} therefore removes the amplification that carries the computation; regularizing ρ (Yang et al. 2026) constrains only the asymptotics, which is exactly the part that should be controlled. Two refinements from direct measurement: first, random-probe transient gain stays ≤ 1 for the recipe even though $\sigma_{\text{max}} \approx 5$, so the amplified subspace is thin: random perturbations decay monotonically, and computation lives in a low-dimensional amplifying cone inside a globally contracting map. (Worst-case gain requires directed probes; random probes measure the typical case, a methodological point we make explicit.) Second, non-normality without convergence control does not help: Neural GPU reaches $\sigma_{\text{max}}/\rho=17.7$, drifts, and scores zero. The recipe’s role is to put strong non-normality *inside* a settling envelope.

Scope of the estimates

Beyond the instruments named in the main text, the per-run inventory includes trajectory length, Koopman/DMD dominant frequencies, a Kolmogorov–Sinai entropy proxy, and gradient statistics at the five training checkpoints; trajectories are stored PCA-reduced to 64 components (metrics use full $d=256$ latents). We state what each quantity is, so it is read for no more than it measures. The Lyapunov “spectrum” is the top eight finite-time exponents (Benettin QR on held-out inputs), not the full $d=256$ spectrum: “all exponents negative” is a statement about those eight, and a Kaplan–Yorke dimension of eight is censored at that cutoff, not an attractor dimension. Eigenvalue-based quantities (Henrici index, numerical and pseudospectral radius) are computed on projected blocks and read as sampled, not exact, normality. $\sigma_{\text{eff}} = \exp(\lambda_{\text{max}})$ is a re-expression of the Lyapunov estimate, not independent corroboration of it. Topological quantities are computed on PCA-reduced trajectories (median 98.5% variance retained). None of these change the regime separations, which are large; they bound the adjectives, not the

verdicts.

The regime-label threshold $|\lambda_{\max}| \leq 0.05$ is likewise not load-bearing: relabeling the corpus at thresholds 0.02, 0.05, and 0.10 orders the median depth conversion of the three regimes identically. As a corpus-level prediction detail, the Lyapunov spread’s AUC 0.846 for extrapolation compares with 0.390 for $|\lambda_{\max}|$ alone: extrapolators are not the most contracting models.

F Full leave-one-out table

Table 3 extends the main text’s Table 1 with every recipe ablation (none of the six additional arms switches dynamical mode; they degrade gracefully) and the SE-RRM baseline (fails the competence gate everywhere it was run). Column definitions are identical to the main text.

G Huginn-3.5B: measurement details

R2 within-family detail. On the last-digit ladder the median freezing step is 12/12/10/12 across operand sizes (1–4 digits): stabilization does not track difficulty *within* a family either, not only across them (main text Sec. 7). On the chain ladder (first rung within competence) accuracy decays 0.72/0.40/0.23 while the freezing step stays put and plateaus by $r \approx 8$ –16.

Probe scope. The behavioral probes are small (304 prompts) and test the R1–R4 criteria, not model ranking; the base model is prompted zero-shot, so absolute accuracies are lower bounds. Across the 2,585 per-token trajectories the effective dimension still tracks task category (2.23 $\rightarrow$ 2.68, trivial to multistep), so the iteration is not inert; the bounded, orbit-like trajectories are consistent with what the model’s authors report (Geiping et al. 2025).

The layer-normalization objection, in full. Is $\sigma_{\text{eff}} = 1.000$ not simply what layer normalization enforces in a pre-norm transformer? Partly, yes, and we lean on it accordingly. Normalization pins the radial direction of the state, so a Lyapunov estimate near zero is partially expected *regardless* of what the map computes, and we do not rest the verdict on it. The verdict rests on trajectory *displacement*, which normalization does not fix: an operator settling toward a fixed point *on* the norm shell would show per-step displacement decaying to zero, and none of the 2,585 measured token trajectories does: the latent keeps traveling at an undiminished rate while the decoded answer sits frozen (answer freezing at median 3–5 changes; main text Sec. 7). This is exactly the premise of Proposition 1(b): a frozen answer with non-vanishing displacement carries no depth-safety guarantee. Normalization explains why marginality is the *default* for this architecture family; it does not explain why training never leaves it; our small-scale constructive experiment (main text Sec. 5) shows a single objective term suffices to leave it when the objective asks.

H Repairing depth-safety at scale: protocol, transfer laws, negative arms

All runs in this section were pre-committed before execution (protocol files in the repository); every result file encodes its

full configuration.

The base pathology. On procedural single-token probes the pretrained model’s accuracy peaks at a shallow, task-dependent recurrence ($r^* \in [2, 24]$; carry ≈ 8 , digit-copy 2, index-digit 24) and then collapses: carry 0.31 ($r=4$) $\rightarrow$ 0.69 ($r=8$) $\rightarrow$ 0.00 ($r \geq 32$), copy dead from $r=4$, letter-repeat 0.56 $\rightarrow$ 0.00. At depth every input freezes onto one input-independent wrong token, so the failure is a settling event (Proposition 1b). Consequently answer-agreement halting cannot detect it: stopping when the decoded answer stops changing recovers 0.56 on carry against 0.74 for an oracle fixed depth, and always-deep inference yields 0.00; a 30-example calibration of a fixed depth recovers the peak.

Repair objective. A rank-16 LoRA adapter on the recurrent core is trained with the single loss $\|z(x, r) - z_{\text{base}}(x, t^*)\| / \|z_{\text{base}}(x, t^*)\|$ on the answer position, $r \sim U[16, 128]$, teacher $t^*=8$, learning rate 10^{-5} , 2,000 steps, no task labels; anchors are two synthetic domains (smallest-digit, first-letter). Held-out results over 10 seeds: digit-copy 0.00 $\rightarrow$ 1.00 ± 0.00 at $r=128$ (all seeds input-dependent by prediction entropy); letter-repeat 0.00 $\rightarrow$ 0.51 ± 0.13 ; carry tier 8 plateau 0.60 ± 0.04 (tiers 2/16: 0.48/0.53) with largely input-independent answers, which we count as stabilization, not restored computation. An 11-domain anchor variant reaches carry 0.68 ± 0.10 (4/9 seeds input-dependent) and holds copy at 0.93 when evaluated at $r=256$; the two-domain recipe is verified to $r=128$. A single first-letter anchor gives the best carry plateau we observed (0.60–0.96 deep; two seeds, one clearly input-dependent).

Transfer follows the anchors’ operation, across symbol alphabets. A single word-morphology anchor (pluralization, which echoes its stem) restores copying in digits (0.98 ± 0.00) and letters (0.89 ± 0.08), input-dependent in 6/6 seeds; carrier cross-tests are diagonal (smallest-digit repairs digit-copy 1.00 but letter-repeat only 0.06; first-letter repairs letter-repeat 0.45 but digit-copy 0.25). Non-echo anchors restore nothing: a fact-association anchor with a stronger base peak than pluralization’s (0.27 vs. 0.17) repairs 0.00, and random-string corpora likewise, so the operation, not anchor strength, carries the transfer.

Two failure laws, measurable in advance. (i) Anchor validity: domains the base itself cannot perform poison the mix. Adding four counting domains (base near-chance) to the two-anchor recipe drops copy from 1.00 to 0.50 ± 0.15 ; an association domain the base scores 0.00 on drives the pluralization anchor’s copy transfer from 0.98 to 0.00. A 30-example base scan of each candidate anchor gates this before training. (ii) Mirror interference: a near-duplicate template of an anchor with contrary semantics is damaged by state capture, not semantic capture: with a smallest-digit anchor, largest-digit falls $0.58 \rightarrow 0.06$ –0.09 deep (90% of its deep answers are not digits at all), and the damage tracks the twin’s share of the training mix; last-letter’s peak falls $0.13 \rightarrow 0.00$ under a first-letter anchor while a non-twin anchor keeps it at 0.14. With no twins among the anchors, no evaluation task is damaged.

	Sudoku (hole fraction)			Reachability (path length)			Addition (carry length)		
	learn	extrapolate	overthink	learn	extrapolate	overthink	learn	extrapolate	overthink
CoRE (full recipe)	1.00	0.19→0.34 ±0.04	0	0.98	0.92→0.92 ±0.10	0	0.99	0.13→0.17 ±0.15	0.05
– structure	0.00	0.00→0.00	0	0.15 ±0.02	0.08→0.08 ±0.03	0	<i>= CoRe (no spatial structure on 1-D)</i>		
– extra input injection	0.29 ±0.01	0.00→0.00	0	0.23	0.17→0.17	0	0.04 ±0.02	0.00→0.00	0
– fixed-point objective	1.00	0.01→0.03 ±0.01	0.04	1.00	0.97→0.97 ±0.02	0.37	1.00	0.14→0.15 ±0.02	0.81
– orthogonal init	1.00	0.19→0.34 ±0.03	0	1.00	0.96→0.96 ±0.01	0.05	1.00	0.18→0.32 ±0.28	0
– adaptive step	1.00	0.11→0.24 ±0.17	0	0.69 ±0.03	0.43→0.45 ±0.18	0.11	1.00	0.39→0.40 ±0.06	0.01
– deep supervision	1.00	0.13→0.37 ±0.04	0	0.99	0.93→0.94 ±0.02	0.07	0.99 ±0.01	0.10→0.27 ±0.22	0.01
– grad clip	1.00	0.10→0.31 ±0.10	0	0.57 ±0.15	0.34→0.34 ±0.19	0.05	0.99 ±0.01	0.08→0.12 ±0.04	0.62
lr $3\cdot 10^{-4} \rightarrow 10^{-3}$	1.00	0.21→ 0.42 ±0.01	0	1.00	0.98→ 0.98 ±0.01	0	0.99 ±0.01	0.27→0.58 ±0.37	0.03
answer+latent objective	1.00	0.20→0.34 ±0.04	0	1.00	0.95→0.95 ±0.02	0.01	1.00	0.07→0.11 ±0.04	0.03
generic recurrence	0.00	0.00→0.00	0	0.17 ±0.01	0.07→0.11 ±0.02	0.13	1.00	0.11→0.12 ±0.02	0.84
TRM-style	0.94 ±0.18	0.14→0.21 ±0.11	0	0.70 ±0.17	0.55→0.56 ±0.22	0.01	1.00	0.80→0.82 ±0.15	0.02
UT/URM	0.97 ±0.06	0.00→0.00	0.18	0.43 ±0.09	0.28→0.28 ±0.07	0.09	0.99 ±0.02	0.66→0.72 ±0.21	0.59
FPRM	0.53 ±0.40	0.00→0.01 ±0.01	0.04	0.31 ±0.05	0.20→0.20 ±0.03	0.11	0.82 ±0.26	0.27→0.28 ±0.25	0.80
EqR	1.00	0.01→0.01 ±0.01	0.13	0.99 ±0.01	0.88→0.89 ±0.03	0.24	1.00	0.12→0.12 ±0.03	0.72
DEQ	1.00	0.00→0.00	0	0.26 ±0.04	0.16→0.16 ±0.03	0.24	0.99 ±0.01	0.19→0.47 ±0.19	0.07
Neural GPU	0.00	0.00→0.00	0	0.76 ±0.08	0.54→0.54 ±0.11	0.68	—	—	—
SE-RRM	0.00	0.00→0.00	0	—	—	—	—	—	—

Table 3: Full leave-one-out and architecture table (mean±sd; architecture rows $n=10$ seeds, CoRE/–fixed-point/TRM-style $n\approx 20$, other ablations and SE-RRM 3); column definitions as in main text Table 1. **learn**: in-distribution EM at best depth. **extrapolate**: first-unseen-tier EM at training depth → best over $h_c \leq 128$. **overthink**: worst over tiers of best EM – EM@128 (bold ≥ 0.2); “—” marks arm undefined or not run.

What never appears: conversion. No treated model exceeds the base’s peak competence at any depth, in any arm. The depth-hard pointer-chase task (iterated single-cycle permutation, $S=10$) is unlearnable in-distribution under LoRA rank 16–128 and full-core tuning at learning rates $3\cdot 10^{-6}$ – 10^{-4} (accuracy at chance, and never increasing with r). The fixed-point loss itself does not transfer to this scale: trained alongside the task it merely freezes an already-frozen loop, and trained alone it collapses the model to a constant answer at every dose we tried, while as little as 50 gentle steps already flatten the base’s depth profile. Where a trained arm does show settling together with high flat accuracy (e.g., a task-trained control at 0.84 on carry with displacement 10^{-5}), the answers are identical across tiers and depths, an input-independent constant rather than competence. Behavioral distillation from the base’s peak depth (a per-domain teacher) repairs a seven-task battery in-mix (5/5 pre-committed criteria; the per-domain teacher matters for copy 0.65 vs. 0.00 and largest-digit 0.62 vs. 0.41 against a uniform teacher) but never holds the deep tail on held-out domains; only the frozen latent anchor transfers.

Seed variance and a label-free selector. Depth-contract training inherits the model’s stochastic latent initialization; where seeds vary (the 11-domain variant), degenerate seeds are detectable without labels by prediction entropy at depth (top-class fraction ≥ 0.85): the selector keeps 9/10 seeds on copy (mean 0.90, dropping exactly the dead seed). The two-anchor recipe required no selection (10/10).

I Weak-form R4: the full natural experiment

The size-ladder experiment separating the two forms of answer convergence (main text Sec. 5): on addition-by-length the generic recurrence length-generalizes ($0.83 \rightarrow 0.86$; a

size-axis result, not an R1 claim) with an answer frozen within ~ 5 steps, *while its latent never settles* ($\sigma_{\text{eff}}=1.02$, still moving at the last step): the carry front travels through the state in directions the readout ignores (weak-form R4). On prefix sums the *same* operator statistics ($\sigma_{\text{eff}}=1.02$, $\lambda_{\text{max}}=+0.02$, spectrally indistinguishable from the addition twin) fail R4: the generic net learns the training tier better than the recipe (1.00 vs. 0.23 at depth 16) and is then destroyed by test-time depth ($1.00 \rightarrow 0.65$ on the same tier, $0.56 \rightarrow 0.04$ on the next), while the recipe’s weaker solution is depth-stable ($0.23 \rightarrow 0.26$).

Strict R2 on a deterministic oracle. The TRM-style model’s halting on addition-carry tracks the oracle’s carry length nearly one-to-one: 16-round carry chains stop at 14.5–15.1 solve steps (main text Sec. 5).

J The fixed-point objective’s gate does not cheat

The fixed-point penalty is scaled by the learned update gate a_T , so the gate could in principle satisfy the penalty by closing ($a_T \rightarrow 0$) without the map learning a fixed point. Trained recipes do not do this: they keep a small but nonzero terminal update and difficulty-adaptive solve steps, unlike the flat-at-2.0 solve steps of the identity mode (Sec. L).

K Objective interventions: dose, matched control, beyond-horizon

These experiments support the causal analysis of main text Sec. 5.4; protocols were committed before the runs (Sec. R).

Answer-stabilization control (isolating the latent attractor). We train the recipe with the residual penalty redirected from the latent to the *decoded answer* (penalizing

$\|\text{readout}(z_T) - \text{readout}(z_{T-1})\|$, leaving the latent free). This freezes the output without asking the latent to converge. It reproduces neither payoff: on Sudoku it fits its easy training tiers (hardest train tier only 0.18–0.30) and extrapolates to 0.00 on all three seeds, with the latent never settling ($\lambda_{\max}=+0.03\text{--}0.07$); on reachability it matches the recipe’s unseen-tier peak (0.91–0.95) but is then destroyed by depth (worst-tier overthinking damage 0.99–1.00, versus 0.00 for the latent-settling recipe). The payoff therefore requires the latent fixed point the objective creates, not answer freezing.

Dose-response of the objective weight. Sweeping the objective coefficient (three seeds each; the headline weight is 0.05): *carry propagation, generic recurrence*: at weight 0 the operator drifts ($\lambda_{\max}=+0.04$), with depth damage 0.87–0.93 and conversion only 0.09–0.14; at 0.005 the exponent crosses into settling and the damage vanishes (0 of 3 unsafe) while conversion is still small; at 0.02 conversion appears (0.60–0.97); at 0.05–0.2 it saturates (≈ 0.88 , matching the headline; 3-seed dose points). Adding the objective thus fixes both the damage and the missing conversion; depth-safety switches on at the settling crossing, and conversion then grows with weight.

Sudoku, recipe: weights 0.01 and 0.05 are comparable (extrapolation 0.32–0.41), weight 0.2 suppresses it (0.05–0.16), and weight 1.0 over-contracts into a deep settle ($\lambda_{\max}=-0.26\text{--}-0.36$) with perfect training accuracy but zero extrapolation. Settling is necessary but not sufficient: the useful target is the slowest rate that still settles.

EqR, native weight: the same over-dose boundary appears architecturally. EqR hard-wires this penalty at weight 0.1, yet on these ladders its residual never resolves (the training loss plateaus at the penalty term while in-distribution EM is perfect), and the model overthinks instead of converting (Table 3). With the native term removed, the unchanged backbone converges and converts (carry $0.68 \rightarrow 1.00 \pm 0.01$, Sudoku $0.14 \rightarrow 0.34 \pm 0.04$; $n=10$ each), the trained operator settling on its own on carry. The objective pays only where the penalty can actually be driven down; a hard-wired weight over a residual that stays large buys neither safety nor conversion.

Constructive transfer across architectures. Re-seeded to $n=10$ per cell, adding only the fixed-point term (weight 0.05) to the two marginal architectures flips the measured regime seed-by-seed: on carry and reachability every UT/URM and FPRM control run has $\lambda_{\max} > 0$ and every +fp run $\lambda_{\max} < 0$ (40/40 runs), and overthinking is repaired wherever there is competence to protect (carry damage $0.59 \rightarrow 0.17$ for UT/URM, $0.80 \rightarrow 0.14$ for FPRM). Conversion does not come along automatically: it rises on FPRM ($0.28 \rightarrow 0.49$) but falls on UT/URM ($0.72 \rightarrow 0.54$), and on reachability the term costs trainability itself (learn $0.43 \rightarrow 0.18$ and $0.31 \rightarrow 0.13$, below the competence gate). DEQ marks the other boundary: its control already settles on carry ($\lambda_{\max} < 0$, 10/10), and adding the term stops training from fitting at all (10/10 seeds). Under the intervention the regime and depth-safety move together on foreign architectures, so the arm’s identity does not carry them; conversion stays task- and architecture-dependent, as in the main text (settling is

necessary, not sufficient).

Beyond-horizon conversion on a deterministic oracle.

To test conversion strictly past the training horizon we re-train at $T_{\text{train}}=8$ and evaluate on oracle depths that exceed it (carry length up to 28, i.e. $3.5 \times$ the horizon). Across 25 seeds the at-horizon tier (carry-8) converts on 17 (mean EM 0.59), but the strictly-beyond-horizon tiers do not convert: the 16-round tier exceeds EM 0.5 on 1 of 25 seeds (mean 0.06) and the 28-round tier on none (mean 0.02). An initial five-seed run had a single seed convert the deep tiers (0.88/0.46); this did not replicate at scale, so we make no strictly-beyond-horizon conversion claim on carry. What is robust strictly past the horizon is depth-safety: the settling recipe holds a flat plateau where its no-fixed-point twin peaks higher and then collapses (the safety contrast is sharper than the conversion contrast, as at $T_{\text{train}}=16$).

L Failure-mode portraits

Full quantitative portraits of the four failure modes (main text Sec. 5):

The identity mode (no-recall): the loop converges instantly to “do nothing” ($\sigma_{\max}=1.0000$; solve steps 2.0 flat, the only arm with no difficulty-adaptivity at all; rank curve frozen). R2–R4 all fail in the most degenerate way; accuracy 0.00.

The marginal mode (generic recurrence, UT/URM, FPRM, SE-RRM): the trained operator is close to normal and norm-preserving (Henrici departure 0.000, $\sigma_{\max}=\rho=1.000$), with a near-zero-to-weakly-positive $\lambda_{\max}$ sitting at or just above the marginal band (values around $+0.05\text{--}0.07$ straddle the marginal/drift threshold: these operators are the least settled non-drift family, not cleanly interior to marginal), saturated Kaplan–Yorke dimension, and positive entropy proxy. Latents wander a high-dimensional filament (trajectory effective dimension ≈ 1.1 , path length $\sim 15k$); rank curves stay high and non-monotone. On the original ladders these models fit easy tiers (UT/URM reaches 0.56 training-tier EM on those tasks; on the fixed-size ladders of Table 1 the same family reaches 0.97; the two figures are different corpora, Sec. B) but R1 fails: their depth curves are flat or declining.

The drift mode (no fixed-point objective): non-normality survives ($\sigma_{\max}=5.6$) but $\lambda_{\max}=+0.10$ ($\sigma_{\text{eff}} 0.94 \rightarrow 1.11$ relative to the recipe); the trajectory is ten times longer than the recipe’s, entropy is the highest measured, and extra test-time steps actively hurt (R4 fails with consequences for R1: $0.33 \rightarrow 0.04$).

The settle-but-wrong mode (no-structure; TRM on ARC): the dynamics are right, as no-structure contracts hardest of all arms; TRM on ARC settles with $\lambda_{\max}=-0.21$ and $\sigma_{\max}=30\text{--}60$, but the fixed point does not encode the solution (EM 0.00 and 0.013 respectively). Convergence control cannot substitute for an inductive bias aligned with the algorithm (main text Sec. 4). Spectrally these runs settle *isotropically* (Lyapunov spread ≤ 0.08), whereas true algorithm-learners settle *anisotropically* (spread 0.2–0.4); the instrument that separates this mode from success is developed in main text Sec. 6.

M Practitioner’s dashboard

Table 4: what to monitor while training a recursive model, what each pathology looks like, and which recipe component treats it (main text Sec. 6).

Signal	Diagnosis	Fix
$\sigma_{\max} \rightarrow 1.0000$, solve steps flat	identity fixed point (“do nothing”)	extra input injection (recall)
Henrici $\rightarrow 0$, $\sigma_{\max}=\rho=1$, KY saturated	normal marginal wander (coverage learner)	structured step; fixed-point objective
$\lambda_{\max}>0$, trajectory length exploding	drift; depth will hurt at test time	fixed-point objective
settles but EM = 0; flat Lyapunov spectrum ($\lambda_1 - \lambda_8 < 0.1$)	settled isotropically; converged to a non-algorithm	inductive bias (structure) / less augmentation, not more dynamics control
all eight estimated leading exponents < 0 , rank curve monotone $\downarrow$, EM rises with depth	learned recursion	none needed
seed-to-seed EM variance $\gg$	missing step-size adaptation / clipping	adaptive updates; clip

Table 4: Signal $\rightarrow$ diagnosis $\rightarrow$ fix: what to monitor while training a recursive model, the pathology each pattern indicates, and the recipe component that treats it.

N Corpus regularities: full statements

One settle axis, measured two independent ways. Live spectral measurements and post-hoc trajectory geometry agree run-by-run: σ_{eff} correlates with trajectory settling at $r=0.75$ ($n=325$, $p=4 \times 10^{-60}$) on the original campaign and replicates on the fixed-size-ladder corpus as a rank correlation (Spearman $\rho=0.69$, $n=1,206$; rank-based because a few deeply-settled runs distort a Pearson estimate).

Regime statistics. Across the 304 original-ladder runs the 19 extrapolating runs on constraint-propagation tasks separate from the rest on both dynamical axes: $\lambda_{\max}$ (median -0.047 vs. $+0.009$, $p=7 \times 10^{-6}$) and Kaplan–Yorke dimension (0 vs. 8, $p=2.6 \times 10^{-5}$).

Topology. A settled cloud is compact, a drifting one a filament: the near-zero-persistence Betti-1 bar count is positive for 92% of settle-regime trajectory files versus 24% marginal and 2% drift. Within the same run, correct examples live on lower-dimensional, more compact trajectories (Sec. P).

Formation caveats. Mid-training hidden-state stable rank predicts final extrapolation at AUC 0.774, but arm identity partially confounds, so it is a monitoring candidate, not a law; and we do not claim the canonical delayed test-accuracy jump of grokking (the checkpoint grid is too coarse). See also Sec. O.

Anisotropy: instrument-level exception. The no-adastep arm extrapolates with a compressed Lyapunov spectrum, since adaptive update scaling may deform the measurement, so the anisotropy claim is read as a corpus regularity, not a per-run guarantee.

History in the spectrum. Isometrically initialized models end with double-to-triple the weight stable rank of their ablation at indistinguishable accuracy so persistent spectral differences can encode history, not just competence.

O Training-time formation

Figure 7: hidden-feature rank across the five training checkpoints. Arms that will extrapolate grow rank late; the no-fixed-point ablation finds its shortcut early and stays flat (main text Sec. 6). This is the inference-map face of delayed generalization: as in grokking (Power et al. 2022), the generalizing solution forms long after the loss is low, visible in hidden progress measures before behavior (Nanda et al. 2023; Barak et al. 2022), competing with a cheaper memorizing solution (Varma et al. 2023), the coverage solutions of the main text. The quantities a grokking analysis would monitor on the training map (rank, spectral structure) are exactly the ones that diagnose the trained inference map. Within the same runs, correct examples additionally live on lower-dimensional, more compact trajectories ($p=3.5 \times 10^{-10}$ and $p=2.3 \times 10^{-6}$ within-run; Sec. P).

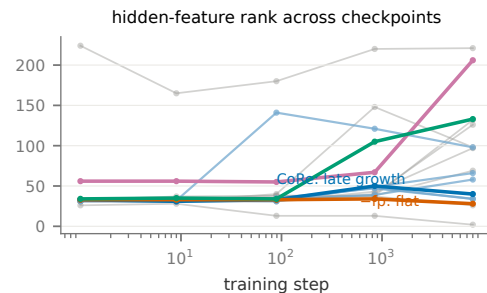


Figure 7: Hidden-feature rank across training checkpoints (Sudoku; colors as in the main text).

P Per-instance confidence read-out

The per-instance predictor of the main text (Sec. 6) is a logistic read-out of five per-trajectory quantities (settling ratio, effective dimension, path length, winding, Koopman magnitude) with grouped cross-validation over runs (no run leaks across folds). Cross-validated AUC: 0.80 (Sudoku), 0.91 (reachability), 0.98 (size-Sudoku); correct examples settle more, on shorter, lower-dimensional paths ($p=3.5 \times 10^{-10}$ and $p=2.3 \times 10^{-6}$ within-run).

Q Data generation

Every generator is deterministic given a seed, verifies its oracle-round invariant on each instance at generation time, and materializes splits once that are shared byte-for-byte

across arms. We describe the three primary fixed-size ladders; the remaining generators follow the same contract (fixed length, vocabulary, and grid across a ladder, with only the oracle-round count growing) and are included in the code supplement.

Sudoku (hole fraction). For each instance we draw a complete valid 9×9 solution by randomized backtracking, then blank a fraction h of the 81 cells chosen uniformly at random ($\lfloor 81h \rfloor$ cells set to the blank token). The givens are exactly the un-blanked solution cells, so every puzzle is consistent with its solution by construction. Difficulty is the hole fraction h (training 0.2/0.35/0.5, evaluation 0.2–0.7). Uniqueness is not enforced at generation: we compute the uniquely-solvable fraction per tier (1.00/0.91/0.50/0.06/0.00) with an exhaustive solver and score predictions with the validity oracle, reporting reference match alongside. The deterministic-oracle ladders below have no such ambiguity.

Reachability (BFS depth). Undirected graphs on $N=16$ nodes with a fixed reachable set of $R=13$. From a source we build d non-empty layers, place at least one edge from each node to the previous layer, and add adjacent-layer edges with probability 0.25, so the BFS depth of every reachable node equals its layer index and the farthest is exactly d . The remaining $N - R=3$ nodes form a separate component (internal edge probability 0.4), so the reachable fraction R/N is constant across tiers and carries no base-rate signal. Node identities are permuted at random, and each instance is verified (BFS depth = d) before it is kept. Difficulty is the exact BFS depth d (training 2, 3; evaluation 2, 3, 5, 8, 12). The input is the flattened adjacency with the source marked on its diagonal; the target is the per-node reachable bit.

Addition (carry-run length). Two 32-bit operands in an interleaved least-significant-bit-first encoding ($L=64$; sum bit i at position $2i$, odd positions padding), which keeps the carry chain strictly local. Difficulty is the longest run c of consecutive live carries: we plant one generate-then-propagate cascade of length exactly c and absorb any accidental longer run, verifying c by carry simulation. The oracle-round count equals c (a local ripple needs c sequential steps) and is recomputable from the input. As pre-committed, the fraction of propagate columns grows with c , as the hole fraction does along the Sudoku ladder.

R Reproducibility details

Every configuration trains at 16 recurrent steps with deep supervision (4 supervised unrolls), AdamW at learning rate 3×10^{-4} , gradient clipping 0.5, batch 64, hidden width $d=256$, 800 unique unaugmented examples per difficulty tier, and three seeds; evaluation uses 256 fixed examples per tier and sweeps test-time depth $h_c \in \{3, 16, 32, 64, 128\}$. Data files are materialized once and shared byte-for-byte across every arm and architecture. Table 5 lists, per task, the difficulty parameter, the training and evaluation tiers (as parameter values), and the training-step budget. Eight runs fell back to a halved batch after an out-of-memory event and continued from there; the actual batch size is logged per run, so no run silently changes budget. By “pre-committed” we mean

that each campaign’s protocol, arm list, and headline choice were committed to version control before the corresponding run and fixed there rather than chosen after inspecting results; we say “pre-committed” rather than “pre-registered” because the commits are not timestamped by a third party.

Availability. An anonymized code supplement accompanies the submission: generator code and materialized splits with checksums, exact seeds and per-arm configurations, the measurement and estimator code with the numerical settings of Sec. E, architecture implementations, per-run and per-example logs, and scripts that regenerate every table and figure. It also lists the eight runs that fell back to a halved batch after an out-of-memory event, with their effective budgets.

task	difficulty	train	eval	steps
<i>Fixed-size ladders (depth axis; primary)</i>				
sudoku	hole fraction	0.2,0.35,0.5	0.2–0.7	8k
reach_pathlen	BFS depth	2,3	2,3,5,8,12	4k
addition_carry	carry-run len	2,4	2,4,8,16,28	4k
maze_pathlen	path length	6,12	6,12,24,48,80	6k
sort_disorder	displacement	2,4	2,4,8,16,24	4k
maze_pathlen_m	path length (matched)	8,12	8,12,16,24,32	6k
sort_disorder_m	displacement (matched)	2,4	2,4,6,8,12	4k
<i>Size/length ladders (scope; main text Sec. 3)</i>				
sudoku_size	board size	9	4,9,16,25	6k
reachability	#nodes	8,12	8,12,16,24	4k
maze	grid size	11,15	11,15,21,31	6k
addition	operand bits	8,16	8–128	4k
prefix	bit length	32,64	32–512	4k
parity	bit length	32,64	32–512	4k
sort	length	8,16	8,16,32,64	4k

Table 5: Per-task difficulty parameter, training/evaluation tiers (parameter values), and training-step budget. All other hyperparameters are shared (text above).